

Brain Tumor Classification with Pretrained CNNs in PyTorch

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Abstract

Brain tumors need to be diagnosed quickly and correctly, and manual review of MRI scans by radiologists, while accurate, is slow and can vary from one reader to another. Convolutional neural networks (CNNs) offer a way to automate part of this process and support, rather than replace, clinical judgment. This paper presents a complete and reproducible transfer learning pipeline that sorts brain MRI scans into four categories: glioma, meningioma, pituitary tumor, and no tumor. We fine-tune a ResNet-18 pretrained on ImageNet, replacing its final layer with a four-way classifier, and train on 5,712 images with light augmentation, early stopping, and learning rate scheduling. On a fully held-out test set of 1,311 images the model reaches 99.16% accuracy, a macro-averaged F1 score of 0.991, and a macro-averaged one-vs-rest AUC above 0.999, misclassifying 11 scans in total. We report the complete confusion matrix and per-class metrics rather than a single headline number, analyze every error the model makes, and note that several errors are made with high confidence, which is a calibration problem worth taking seriously in a clinical context. Grad-CAM heatmaps confirm the network attends to brain tissue rather than scan borders or acquisition artifacts. All code, figures, and hyperparameters are released so the results can be reproduced end to end on a single consumer GPU.

1 Introduction

A brain tumor is an abnormal growth of cells inside the skull, and depending on type and location, it can be life-threatening. Tumors are classified as primary, meaning they originate in the brain, or secondary, meaning they spread from cancer elsewhere in the body. Among primary tumors, gliomas and meningiomas are the most common, with gliomas generally being more aggressive. Because symptoms depend heavily on tumor size and location, early and accurate detection matters a great deal. Diagnosis has traditionally relied on radiologists reading MRI scans by hand, a process that takes time and is subject to disagreement between readers. Given how much a delayed or missed diagnosis can affect a patient's outcome, there is real value in tools that can support this process.

Deep learning, and CNNs in particular, has changed how image analysis is done by learning hierarchical features directly from pixels instead of relying on hand-crafted rules. This has made it possible to segment and classify medical

images without manually engineering features for every task. When labeled data is limited, which is common in medical imaging, transfer learning helps close the gap: a CNN pretrained on a large natural-image dataset like ImageNet is fine-tuned on the smaller medical dataset instead of being trained from nothing. Tajbakhsh et al. (2016) showed that fine-tuned pretrained models frequently outperform models trained from scratch on medical imaging tasks. CNNs have since been applied widely to both segmentation and classification of brain MRI. Havaei et al. (2017), for example, combined local and global image context in a single CNN to achieve strong glioblastoma segmentation results. On the classification side, several recent studies report high accuracy with transfer learning: Disci et al. (2023) fine-tuned architectures such as ResNet50 and Xception and reported over 98% accuracy across four tumor categories, while Vimala et al. (2023) used EfficientNet variants to reach roughly 99% test accuracy distinguishing glioma, meningioma, and pituitary tumors.

The contribution of this paper is not a new architecture. It is a clean, fully documented baseline that anyone can rerun and get the same numbers from. Specifically:

- We describe the entire pipeline end to end, from folder layout and preprocessing through training, evaluation, and interpretability, with every hyperparameter stated (Sections 4 and 5).
- We report complete results rather than a headline accuracy: per-class precision, recall, and F1, the raw confusion matrix, per-class ROC-AUC, and an analysis of all 11 test-set errors (Section 6).
- We include Grad-CAM interpretability and use it to check that the model attends to anatomically plausible regions (Section 7).
- We are explicit about what this baseline does not establish, including calibration, generalization across institutions, and run-to-run variance (Section 8).

2 Related Work

CNNs have performed well across a range of medical imaging tasks, including tumor segmentation and detection. Havaei et al. (2017) introduced a fully automatic brain tumor segmentation network that improved both speed and accuracy over earlier methods by combining local and more global contextual

features. Pereira et al. (2016) took a related approach, building a deep CNN for glioma segmentation using small convolutional kernels stacked in depth. These segmentation studies motivate our use of CNNs for classification, since the same hierarchical features that trace tumor boundaries can plausibly distinguish between tumor types.

Transfer learning is a natural fit for classification problems with limited labeled data. Tajbakhsh et al. (2016) studied medical image analysis tasks spanning radiology and cardiology and found that CNNs pretrained on ImageNet and then fine-tuned generally outperform networks trained from scratch, and that fine-tuned networks are also more robust when training data is scarce. MRI scans look nothing like ImageNet’s natural photographs, but this line of work is exactly why we still start from an ImageNet-pretrained ResNet-18: fine-tuning lets the network adapt its low-level filters to MRI texture while keeping the general-purpose pattern detectors learned from a much larger dataset.

Several studies have applied transfer learning specifically to brain MRI. Disci et al. (2023) fine-tuned Xception, ResNet50, and DenseNet121 on a public brain MRI dataset and reported weighted accuracies up to 98.7% across glioma, meningioma, pituitary, and healthy classes. Vimala et al. (2023) optimized EfficientNet-B0 through EfficientNet-B4 on a comparable contrast-enhanced MRI dataset and reported approximately 99% accuracy across three tumor types. Other work looks at combining models rather than picking one: Chen et al. (2024) fused features from ResNet101, DenseNet121, and EfficientNet to improve robustness, reporting over 97% accuracy on standard benchmarks. Taken together, these results suggest that pretrained CNNs, given careful fine-tuning and reasonable data augmentation, can classify brain tumors with high accuracy. What is often missing from these reports is the detail needed to reproduce them exactly, which is the gap this paper tries to fill.

3 Dataset

We use the brain MRI dataset released by Ghaffar et al. (2020) on Mendeley Data, which contains 7,023 T1-weighted, contrast-enhanced MRI images spanning four classes: glioma, meningioma, pituitary tumor, and no tumor. The images are JPEG-formatted axial and coronal slices, gathered from multiple sources and pre-split by the dataset authors into training and testing folders. That split matters for evaluation: our pipeline treats the training folder as the only source of trainable data and keeps the testing folder completely untouched until final evaluation, so no test image influences either the weights or the choice of checkpoint.

Table 1 gives the exact composition of both splits as counted directly from disk. The classes are reasonably balanced, with `notumor` the largest at 1,595 training images and `glioma` the smallest at 1,321. Figure 1 shows example scans from each class, and Figure 2 plots the same counts visually. Both figures are generated by `visualizer.py` from whatever copy of the data is on disk rather than hardcoded, so they always

describe the data actually being trained on.

Table 1: Dataset composition, counted from disk. The training split is further divided 80/20 into 4,569 training and 1,143 validation images.

Class	Training	Testing	Total
glioma	1,321	300	1,621
meningioma	1,339	306	1,645
notumor	1,595	405	2,000
pituitary	1,457	300	1,757
Total	5,712	1,311	7,023

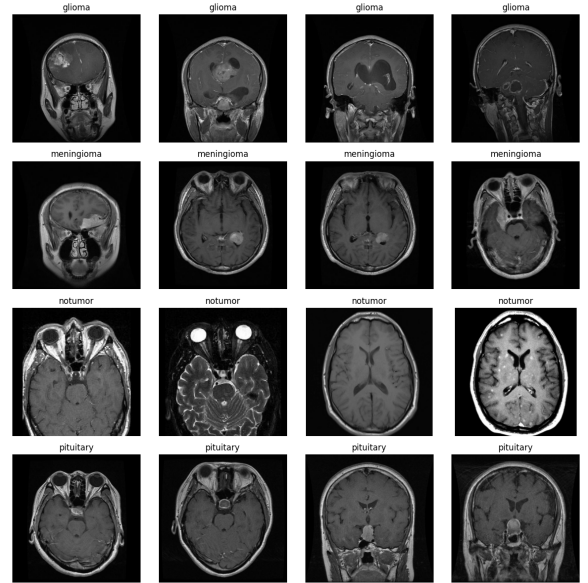


Figure 1: Sample MRI scans from each of the four classes.

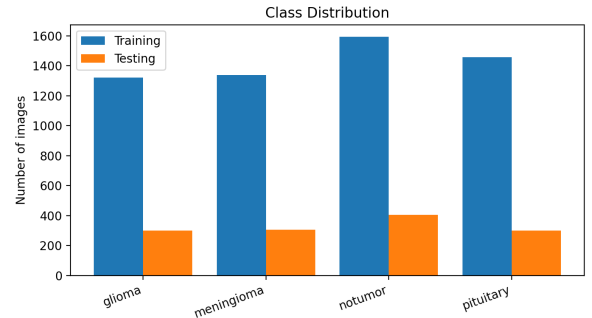


Figure 2: Number of images per class in the training and testing splits, generated by `visualizer.py`.

To match ImageNet-pretrained models such as ResNet-18, every image is resized to 224×224 and grayscale scans are copied across three channels to look like RGB input. Pixel values are then normalized using ImageNet’s mean and standard deviation so the pretrained convolutional filters and

batch-normalization statistics behave as intended. This dataset is used widely for benchmarking brain tumor classification (Disci et al., 2023; Vimala et al., 2023), and its open access and balanced classes make it a practical choice for reproducible research.

4 Methodology

Figure 3 outlines the full pipeline. An input MRI scan is preprocessed to match the format ResNet-18 expects, passed through the pretrained backbone and a newly initialized four-class head, and converted to class probabilities. The resulting predictions feed both the quantitative evaluation and the Grad-CAM interpretability step described in Sections 6 and 7.

Because pretrained CNNs offer a strong balance of representational power and computational cost, we build on ResNet-18 (He et al., 2016) throughout, though the accompanying code also supports ResNet-34 and EfficientNet-B0 as drop-in alternatives.

4.1 Data Preprocessing

Figure 4 shows the preprocessing chain. All MRI images are resized to 224×224 pixels to match ResNet-18’s expected input. Since the source images are grayscale, the single channel is duplicated three times to imitate an RGB image. Pixel values are then normalized using ImageNet’s mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). These steps keep the pretrained model’s early filters and batch-normalization statistics operating in the input range they were trained on. When augmentation is enabled, a random horizontal flip, a rotation of up to 10 degrees, and brightness and contrast jitter of up to 10% are applied before normalization. None of these transformations change the diagnostic label of the scan, which is what makes them safe to use here.

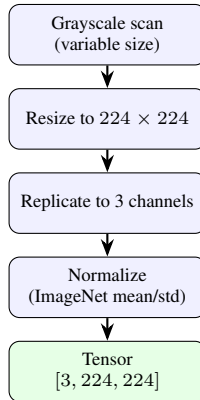


Figure 4: Preprocessing steps applied to every scan before it reaches the network.

4.2 Model Architecture

We use ResNet-18 as implemented in `torchvision.models`, pretrained on ImageNet. Figure 5

shows the layout: the convolutional stem and four residual stages carry over their pretrained weights, and the final fully connected layer, originally sized for 1,000 ImageNet classes, is replaced with a new linear layer with four outputs, one per tumor class. The entire network is fine-tuned rather than just the new head, since full fine-tuning has been shown to outperform freezing early layers on many medical imaging tasks (Tajbakhsh et al., 2016). The code also supports freezing the backbone (`--freeze_backbone`) for faster experimentation on smaller GPUs.

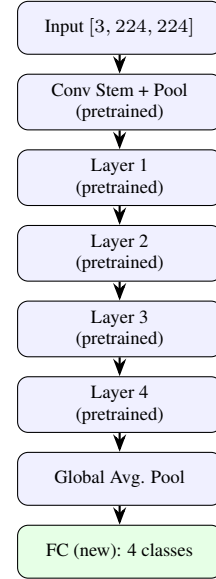


Figure 5: ResNet-18 with its final layer replaced. Blue blocks keep their ImageNet-pretrained weights and are fine-tuned; the green block is initialized from scratch. Layer 4 is also the target layer for Grad-CAM.

4.3 Training Procedure

Training uses cross-entropy loss and the Adam optimizer. The official training folder is split with a stratified 80/20 partition so each class appears in the same proportion in both subsets, giving 4,569 training and 1,143 validation images. A `ReduceLROnPlateau` scheduler halves the learning rate when validation loss stops improving for two epochs, and training stops early if validation loss fails to improve for five consecutive epochs. The checkpoint is saved whenever validation loss reaches a new minimum, so the model that gets evaluated is the best one seen during training and not simply the last one. All random number generators are seeded, and the checkpoint stores the class names and backbone choice alongside the weights, so evaluation and inference never have to guess the class order.

5 Experimental Setup

Table 2 lists every hyperparameter used to produce the results in this paper. Training ran on a single NVIDIA RTX 4070 with

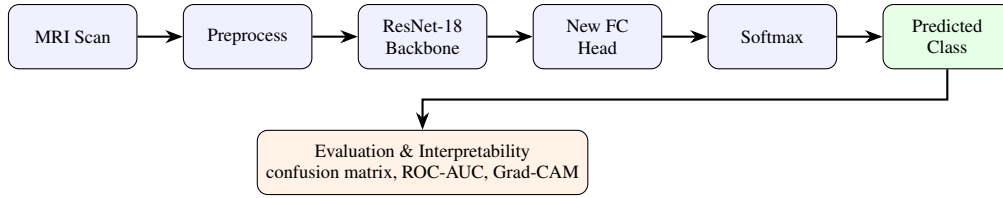


Figure 3: End-to-end pipeline. Preprocessing, the ResNet-18 backbone, and the new classification head are described in Sections 4.1 and 4.2; evaluation and interpretability are described in Sections 6 and 7.

PyTorch 2.14 and CUDA 12.6, and completed in a few minutes; the same script runs unchanged on CPU, just considerably slower. Evaluation metrics come from `scikit-learn`, and every figure in Section 6 is produced directly by the released scripts.

Table 2: Hyperparameters used for the reported run.

Setting	Value
Backbone	ResNet-18 (ImageNet)
Fine-tuning	All layers
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	32
Max epochs	30
Early stopping patience	5
LR scheduler	ReduceLROnPlateau
Scheduler factor / patience	0.5 / 2
Augmentation	Flip, $\pm 10^\circ$, jitter
Random seed	42

6 Results

6.1 Training Dynamics

Figure 6 shows loss and accuracy per epoch. The model converges quickly: validation accuracy passes 98% by the third epoch, which is unsurprising given that the backbone starts from pretrained weights and only has to adapt. After roughly epoch 10 the training loss falls to near zero while validation loss flattens out in the 0.04 to 0.05 range and moves around without trending down. That widening gap is textbook mild overfitting, and it is precisely the situation early stopping exists to handle. Training halted at epoch 21 after five epochs without improvement, and the best checkpoint by validation loss came from epoch 16 (validation loss 0.0376, validation accuracy 99.13%). Peak validation accuracy over the run was 99.21% at epoch 15, slightly higher than at the checkpoint we selected, which is a reminder that selecting on loss and selecting on accuracy do not always agree.

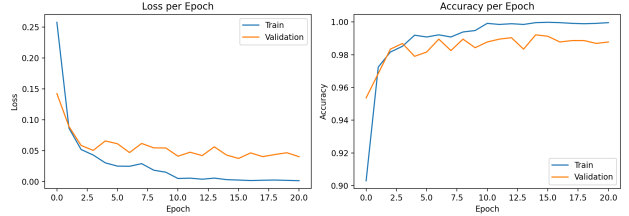


Figure 6: Training and validation loss and accuracy per epoch. Training loss approaches zero while validation loss plateaus, which is what triggers early stopping at epoch 21.

6.2 Test Set Performance

The selected checkpoint was evaluated once on the 1,311 held-out test images. Overall accuracy is 99.16%, corresponding to 1,300 correct predictions and 11 errors. Table 3 breaks this down by class.

Table 3: Per-class test set performance on 1,311 held-out images.

Class	Prec.	Recall	F1	n
glioma	0.997	0.977	0.987	300
meningioma	0.981	0.994	0.987	306
notumor	0.993	0.998	0.995	405
pituitary	0.997	0.997	0.997	300
Accuracy		0.9916		1,311
Macro avg	0.992	0.991	0.991	1,311
Weighted avg	0.992	0.992	0.992	1,311

Performance is consistent across classes, with F1 scores between 0.987 and 0.997. The weakest class is glioma, whose recall of 0.977 is noticeably below its precision of 0.997. In plain terms, when the model says a scan is a glioma it is almost always right, but it misses about 2% of actual gliomas. Given that gliomas are the more aggressive tumor type, a missed glioma is the most costly error in this matrix, so the asymmetry is worth naming rather than averaging away.

6.3 Confusion Matrix and Error Analysis

Figure 7 shows the normalized confusion matrix, and Table 4 gives the raw counts, which are easier to reason about when the errors are this few.

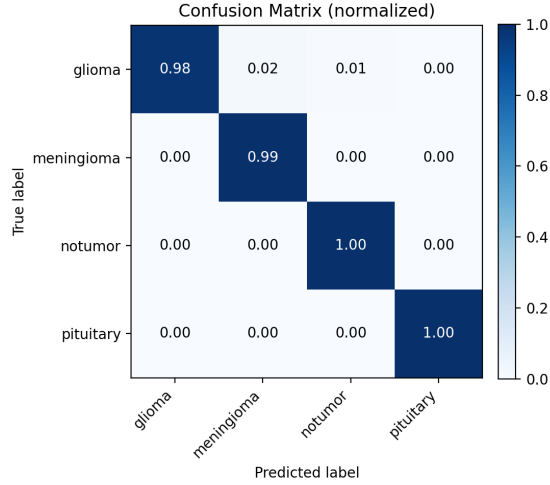


Figure 7: Test-set confusion matrix, normalized by true class.

Table 4: Raw confusion matrix. Rows are true classes, columns are predictions.

True \ Pred	glio.	men.	not.	pit.
glioma	293	5	2	0
meningioma	0	304	1	1
notumor	1	0	404	0
pituitary	0	1	0	299

The errors are not spread evenly. Seven of the 11 come from the glioma class, and five of those are gliomas predicted as meningiomas, making glioma-to-meningioma the single dominant failure mode. This is a clinically sensible confusion: both are common primary tumors, both can present with similar contrast enhancement on a single 2D slice, and distinguishing them often depends on location and volumetric context that a single slice does not carry. The remaining errors are scattered: two gliomas read as no tumor, one meningioma each as no tumor and pituitary, one no-tumor scan read as glioma, and one pituitary as meningioma. The two false negatives where a glioma is called healthy are the most concerning cases in the set, even at only two out of 300.

A detail that a headline accuracy number hides is how confident the model is when it is wrong. Of the 11 errors, four were made with softmax confidence above 0.92, including a glioma called meningioma at 0.964. Only two errors fell below 0.60, which is roughly where a confidence threshold might plausibly route a case for human review. A model that fails loudly and confidently is harder to build a safe workflow around than one whose errors sit near the decision boundary, and this is the clearest evidence in our results that calibration deserves attention before anything like clinical use. Figure 8 shows the misclassified scans themselves.

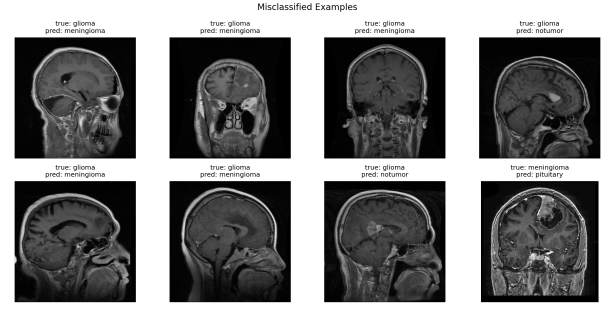


Figure 8: Misclassified test scans with true and predicted labels.

6.4 Ranking Quality

Figure 9 shows one-vs-rest ROC curves. All four classes have an AUC above 0.999 (glioma 0.9999, meningioma 0.9998, notumor 1.0000, pituitary 1.0000), with a macro average of 0.9999. AUC measures how well the model ranks positives above negatives independent of any specific threshold, so these values say the class scores are almost perfectly separable. They should be read alongside, not instead of, the confusion matrix: near-perfect AUC and 11 hard errors at the default threshold are consistent, and together they suggest that a better-chosen operating point could trade some precision for the glioma recall that matters most here.

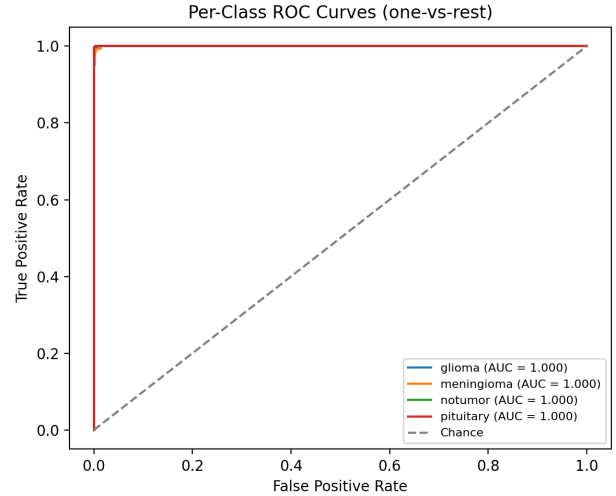


Figure 9: Per-class one-vs-rest ROC curves with AUC.

6.5 Comparison with Prior Work

Table 5 places our result next to recent transfer learning work on brain tumor MRI. We want to be careful about how much weight this carries. These studies use different dataset versions, different splits, and in one case only three classes, so the numbers are indicative of the general performance range rather than a controlled head-to-head comparison. The honest reading is that a plain fine-tuned ResNet-18, which is smaller and cheaper than most of the architectures listed, lands in the

same range as heavier models and ensembles. That is a useful result mostly because it sets a floor: if a more complex method cannot beat this baseline by a clear margin, the added complexity is hard to justify.

Table 5: Reported accuracy in recent related work. Datasets and splits differ, so these are not directly comparable.

Work	Model	Acc.
Chen et al. (2024)	Fusion (3 CNNs)	>97%
Disci et al. (2023)	Xception, ResNet50	98.7%
Vimala et al. (2023)	EfficientNet B0-B4	≈99%
This work	ResNet-18	99.16%

7 Interpretability

A model that predicts well but cannot explain itself is a hard sell in a clinical setting. We include a Grad-CAM implementation (Selvaraju et al., 2017) that highlights the image regions most responsible for a prediction by backpropagating gradients into the final convolutional stage of the backbone (Layer 4 in Figure 5). Figure 10 shows one example per class, all four correctly classified.

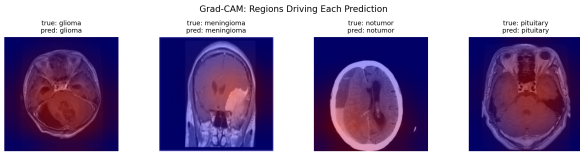


Figure 10: Grad-CAM heatmaps for one example per class. Warmer regions contributed more to the predicted class.

The main thing worth checking with these maps is a negative result: the model is not keying on scan borders, background, or the black region outside the skull, which is the classic failure mode when a dataset aggregates sources with different acquisition conventions and the network learns the source instead of the pathology. Attention falls on brain tissue in all four cases. The meningioma example is the most convincing, with the hottest region sitting over the enhancing lesion itself, and the pituitary example concentrates near the skull base where the pituitary sits.

That said, we would overstate things by calling these tight localizations. The maps are diffuse, often covering a large fraction of the parenchyma, which is expected given that Grad-CAM at Layer 4 operates on a 7×7 spatial grid upsampled to full resolution. Grad-CAM is a sanity check on what the model is looking at, not a segmentation, and a plausible-looking heatmap is not evidence that a prediction is correct.

8 Limitations and Future Work

The results above are strong, and that is exactly why the limitations deserve a fair hearing rather than a paragraph of hedging

at the end.

The most important caveat is that 99.16% on this test set does not mean 99.16% in a hospital. The test split comes from the same curated collection as the training data, with the same sources, scanners, and preprocessing conventions. Performance on data from an institution not represented here would almost certainly be lower, and nothing in these experiments measures that. Multi-institution evaluation is the single most valuable next step.

Second, this is one run with one seed. We report no variance across seeds and no cross-validation, so the difference between 99.16% and a number half a point lower is not something these experiments can resolve. The gap between peak validation accuracy (99.21%) and validation accuracy at the selected checkpoint (99.13%) is itself a hint at how much run-to-run noise sits at this level of performance. Any future comparison between architectures on this dataset should report a mean and spread over several seeds, because at this accuracy the differences between methods are smaller than the noise.

Third, the model is confidently wrong on a meaningful share of its errors, as Section 6 describes. Calibration methods such as temperature scaling, along with proper uncertainty estimates from deep ensembles or Monte Carlo dropout, would be more valuable here than another point of accuracy, since a well-calibrated model can defer uncertain cases to a radiologist and a badly calibrated one cannot.

Fourth, the model reads a single 2D slice and ignores the volumetric structure of the scan. The dominant glioma-to-meningioma confusion is plausibly a direct consequence: the cues that separate those two often live in tumor location and extent across slices. A 3D CNN or a model that aggregates predictions across neighboring slices is the natural fix and is likely to help precisely where this model is weakest.

Fifth, we did not run a hyperparameter search or an ablation study. The settings in Table 2 are sensible defaults from the literature, not the result of a search, and we cannot say how much the augmentation, the scheduler, or full fine-tuning each contributed. Those ablations are cheap to run with the released code and would make the baseline more informative.

Finally, this system only classifies, and it assumes it is handed a usable, roughly centered slice. A real diagnostic tool would need segmentation, automated quality control for artifacts and off-target slices, and integration with patient history to resemble how a radiologist actually works a case.

9 Ethical Considerations

Automated tumor classification raises real ethical questions that should not be glossed over. Tools like this one are meant to support a radiologist’s judgment, not replace it, and this model does make mistakes: it missed two gliomas outright in a test set of 300, and it made several of its errors with high confidence. A false negative can mean a missed diagnosis, and a false positive can mean unnecessary anxiety and follow-up procedures. Any deployment needs a human in the loop and a workflow designed around the assumption that the model will

occasionally be confidently wrong.

Training data should be collected with clear patient consent and proper anonymization, and datasets should be as demographically diverse as the populations a model will actually serve. A model validated only on data like the data in this paper should not be presented as validated for general clinical use. We release this code openly so the methodology can be checked, challenged, and improved by others, which we think is a minimum bar for work aimed at clinical settings.

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